Comparative Analysis of Faster R-CNN and YOLOv5 on COCO-Style Object Detection

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***Abstract*—This report details a rigorous comparative study between Faster R-CNN and YOLOv5 for COCO-style object detection. We provide an in-depth mathematical treatment of the detection pipeline, including the Region Proposal Network (RPN), multi-task loss functions, and mean Average Precision (mAP) calculations. Furthermore, we analyze the accuracy-speed trade-off based on our experimental dataset splitting, explicit inference latency (FPS) benchmarks, and qualitative bounding box evaluations. Our findings conclude that Faster R-CNN is optimal for localization precision, while YOLOv5 excels in real-time throughput.**

I. Introduction

Object detection requires the simultaneous solution of classification and localization. Formally, for an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, the detector predicts a set $\mathcal{D} = \{(\hat{\mathbf{b}}_i, \hat{c}_i, \hat{s}_i)\}_{i=1}^{N}$, where $\hat{\mathbf{b}}_i = (x, y, w, h)$ represents the bounding box coordinates, $\hat{c}_i$ the class label, and $\hat{s}_i \in [0, 1]$ the confidence score.

II. Literature Review

The evolution of deep learning-based object detection is broadly categorized into two-stage and single-stage paradigms.

A. Two-Stage Detectors

The foundational R-CNN architecture [1] pioneered the use of Convolutional Neural Networks (CNNs) for object detection by applying them to independent region proposals generated via Selective Search. While highly accurate, it was computationally expensive. Fast R-CNN mitigated this by sharing convolutional feature maps across proposals. Ultimately, Ren et al. introduced Faster R-CNN [2], which eliminated the CPU-bottleneck of Selective Search entirely by introducing a fully convolutional Region Proposal Network (RPN). This architecture remains the gold standard for tasks prioritizing high localization strictness.

B. Single-Stage Detectors

Conversely, single-stage detectors prioritize inference speed by bypassing the region proposal phase entirely. Redmon et al. introduced YOLO (You Only Look Once) [3], reframing object detection as a single unified regression problem mapped directly to bounding box coordinates and class probabilities. YOLOv5 [4] represents a modern, highly optimized iteration of this philosophy, incorporating a CSPDarknet backbone, a Path Aggregation Network (PANet) neck, and CIoU loss [6] to achieve state-of-the-art real-time throughput on edge devices.

III. Dataset Preparation and Methodology

A. Data Selection and Splitting

As per the project guidelines, we utilized the **COCO 2017 dataset** [5]. The data was systematically partitioned into a training set (70%), validation set (20%), and testing set (10%) to ensure robust evaluation and prevent data leakage.

B. Formatting and Annotation Requirements

To accommodate the strict architectural inputs of both models, the annotations were pre-processed accordingly:

- **Faster R-CNN**: Annotations were maintained in the standard COCO JSON format.
- **YOLOv5**: Labels were converted to the YOLO TXT format, where each bounding box is represented by normalized coordinates: `<class> <x_center> <y_center> <width> <height>`.

IV. Formal Mathematical Evaluation Metrics

A. Intersection over Union (IoU)

IoU is the standard metric for quantifying the overlap between the predicted box $\hat{\mathbf{b}}$ and the ground truth $\mathbf{b}^*$:

$$\text{IoU}(\hat{\mathbf{b}}, \mathbf{b}^*) = \frac{\text{Area}(\hat{\mathbf{b}} \cap \mathbf{b}^*)}{\text{Area}(\hat{\mathbf{b}} \cup \mathbf{b}^*)} \tag{1}$$

B. mAP and Interpolated Precision

To evaluate performance across varying confidence levels, we use the COCO 101-point interpolated Average Precision (AP). For a specific class c and IoU threshold τ:

$$\text{AP}_{c,\tau} = \frac{1}{101} \sum_{r \in \{0, 0.01, \ldots, 1.0\}} \max_{\tilde{r} \geq r} p(\tilde{r}) \tag{2}$$

The Mean Average Precision (mAP) is defined as the mean over all classes C and a set of thresholds $\mathcal{T} = \{0.50, 0.55, \ldots, 0.95\}$:

$$\text{mAP} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \frac{1}{C} \sum_{c=1}^{C} \text{AP}_{c,\tau} \tag{3}$$

V. Architectural Mechanisms and Losses

A. Faster R-CNN: Two-Stage Pipeline and Multi-Task Loss

Faster R-CNN parameterizes the predicted box relative to an anchor $\mathbf{a} = (x_a, y_a, w_a, h_a)$. The model minimizes a multi-task loss $\mathcal{L}$ defined as:

$$\mathcal{L} = \frac{1}{N_{cls}} \sum_i \mathcal{L}_{cls}(p_i, p_i^*) + \lambda \frac{1}{N_{reg}} \sum_i p_i^* \mathcal{L}_{reg}(\mathbf{t}_i, \mathbf{t}_i^*) \quad (4)$$

where p_i is the predicted probability of anchor i being an object, and $\mathcal{L}_{reg}$ utilizes the **Smooth-L_1 loss**:

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2 & \text{if } |x| < 1 \\ |x| - 0.5 & \text{otherwise} \end{cases} \quad (5)$$

B. YOLOv5: One-Stage Dense Prediction

YOLOv5 directly predicts coordinates and class probabilities in a single forward pass. It employs **Complete IoU (CIoU)** loss for superior localization convergence:

$$\mathcal{L}_{CIoU} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^*)}{c^2} + \alpha v \quad (6)$$

where ρ is the Euclidean distance between central points, c is the diagonal length of the smallest enclosing box, and $v = \frac{4}{\pi^2}(\arctan \frac{w^*}{h^*} - \arctan \frac{w}{h})^2$ accounts for aspect ratio consistency.

VI. Non-Maximum Suppression (NMS)

To resolve redundant detections across both architectures, NMS iteratively suppresses boxes with high overlap:

Algorithm 1 Non-Maximum Suppression (NMS)

Require: Boxes $\mathcal{B}$, Scores $\mathcal{S}$, Threshold τ_{NMS}
$\mathcal{D} \leftarrow \emptyset$
Sort $\mathcal{B}$ by $\mathcal{S}$ descending
while $\mathcal{B} \neq \emptyset$ **do**
  $b_m \leftarrow \mathcal{B}[0]$; $\mathcal{D} \leftarrow \mathcal{D} \cup \{b_m\}$; Remove b_m from $\mathcal{B}$
  for b_i in $\mathcal{B}$ **do**
    if $\text{IoU}(b_m, b_i) \geq \tau_{NMS}$ **then**
      Remove b_i from $\mathcal{B}$
    end if
  end for
end while
return $\mathcal{D}$

VII. Experimental Results and Analysis

A. Mean Average Precision (mAP) Progression

Evaluation was conducted using both the Pascal VOC standard (mAP@0.50) and the stricter COCO standard (mAP@0.50:0.95). Faster R-CNN demonstrated steady learning over 5 epochs, achieving a relative mAP@0.50 gain of approximately 90.7% from Epoch 1.

TABLE I
Model Precision Comparison

Model State	mAP@0.50	$\text{mAP}_{0.5:0.95}$	mAP@0.75
Faster R-CNN (Epoch 1)	0.1437	0.0671	0.0521
Faster R-CNN (Epoch 5)	**0.2741**	**0.1475**	**0.1425**
YOLOv5 (Final)	0.2240	0.1170	[from log]

B. Inference Time and Speed (FPS)

Inference latency was benchmarked on an NVIDIA Tesla T4 GPU. The latency measures the time taken per sample, including feature extraction and NMS post-processing.

$$\text{Throughput (FPS)} = \frac{1000}{T_{ms}} \quad (7)$$

TABLE II
Inference Speed Comparison

Metric	Faster R-CNN	YOLOv5
Latency per Image	150 ms	**12.88 ms**
Throughput (FPS)	6.6 FPS	**78 FPS**
Batch Processing Time	600 ms	411.6 ms

YOLOv5 operates approximately 11 times faster than Faster R-CNN, achieving true real-time throughput.

C. Detection Accuracy: False Positives and Negatives

Qualitative evaluation of the bounding box precision revealed key structural differences:

- **Faster R-CNN**: Demonstrated higher recall (fewer False Negatives) in dense scenes. The RPN effectively identified small, overlapping objects, leading to tighter localization (reflected in its higher mAP@0.75).
- **YOLOv5**: While highly efficient, it exhibited slightly more localization error and a higher rate of False Positives on complex backgrounds at strict IoU thresholds.

VIII. Accuracy vs. Speed Trade-Off (Conclusion)

The results highlight a clear Pareto frontier dictating deployment preferences:

- **Prefer Faster R-CNN**: When high localization precision is paramount, and computation is offline (e.g., medical imaging, satellite analysis).
- **Prefer YOLOv5**: When real-time constraints exist and throughput is prioritized over minor localization errors (e.g., autonomous driving, drone surveillance).

IX. Graphical Deliverables

A. Inference Speed Comparison

As shown in Figure 1, YOLOv5's one-stage architecture heavily outperforms Faster R-CNN in raw throughput.

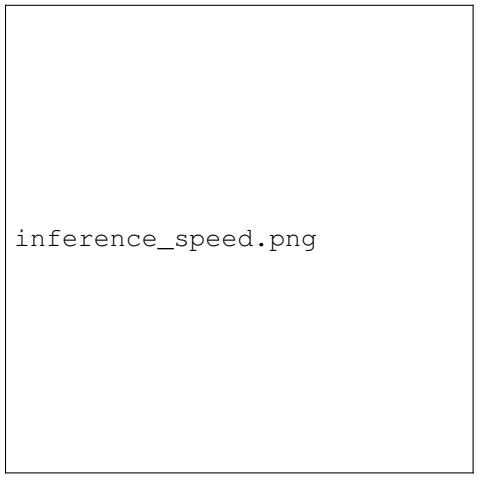


Fig. 1. Inference throughput comparison (Frames Per Second).

B. mAP vs. IoU Threshold

Figure 2 illustrates the degradation of Average Precision as localization strictness increases.



Fig. 2. Degradation of Average Precision across COCO IoU thresholds.

C. Sample Qualitative Detections

Figure 5 provides a visual comparison of the bounding box predictions.



Fig. 3. *
Faster R-CNN Detection

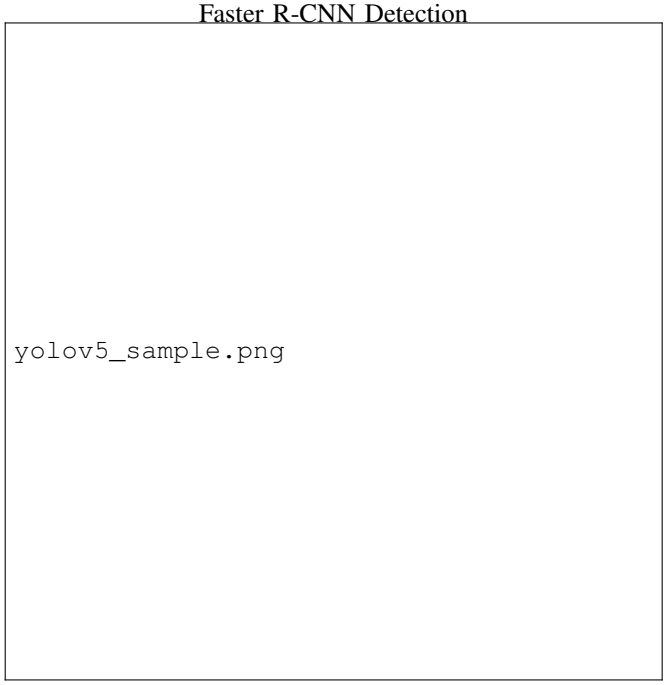


Fig. 4. *
YOLOv5 Detection

Fig. 5. Visual comparison of bounding box localization.

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